

Hierarchical Book Organization for Learning-Resource Discovery using Dual-Path Graph Convolutions

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APPENDIX A DATASET DETAILS

Existing book genre datasets are largely limited to flat classification settings and typically provide only isolated metadata sources such as blurbs, covers, or genre labels. However, literary genre understanding is inherently hierarchical, requiring simultaneous modeling of coarse- and fine-grained genre semantics. Furthermore, prior datasets rarely combine authoritative publisher-provided blurbs with crowd-sourced reader reviews, despite these sources offering complementary perspectives on narrative intent, thematic structure, and reader interpretation. As a result, existing datasets are not well-suited for evaluating hierarchical reasoning over heterogeneous literary evidence. To address these limitations, we introduce a two-level hierarchical genre classification dataset comprising paired blurbs, semantically diverse reviews, and hierarchical genre annotations. The proposed dataset is specifically designed to facilitate research on hierarchical literary genre understanding, multimodal semantic reasoning, and noise-aware integration of authoritative and crowd-sourced textual evidence, which are not jointly supported by existing benchmarks.

Our dataset comprises 9375 books with associated blurb information and 90499 user reviews. At level-1, the books are broadly categorized into two primary categories: *fiction* and *nonfiction*, comprising 5612 and 3763 books, respectively. Each book within these main categories is further annotated with one or more genres at level-2, which includes 28 sub-genres for fiction and 29 for nonfiction. Notably, the level-2 annotations follow a multi-label format. The complete list of level-2 genres is presented in Table A.1.

Building upon the earlier Book-Crossing dataset [1], which often overlooks the inherent hierarchical and multi-label structure of genre taxonomies, we constructed a new dataset by crawling Goodreads (www.goodreads.com). We collected both the blurb and user reviews for each book. While the blurbs were largely consistent and aligned well with the central themes of the books (with only a few instances of missing blurbs), user reviews tended to be noisy and occasionally inconsistent. Furthermore, to annotate the dataset, we began with an initial set of flat genre categories collected from Goodreads, which

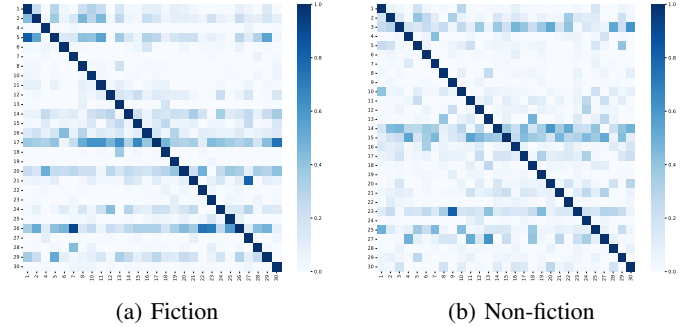


Fig. A.1: Label co-occurrence heatmap of our dataset, where each cell indicates the probability of the column-label occurring under the row-label, indicating the relationship among the labels in the dataset.

often numbered in the thousands. To ensure reliable and high-quality annotations, we recruited five annotators with professional knowledge in English. They underwent rigorous training under the supervision of a linguistics expert to label the data. Their efforts were acknowledged through fair and adequate remuneration. We observe strong annotation reliability, with Fleiss' $\kappa = 0.78$, indicating substantial agreement among annotators. The heat maps in Figs A.1 (a), (b) illustrate the label co-occurrence and sparsity patterns among fiction and nonfiction genres, based on the final annotated dataset.

Following annotation, while developing the genre prediction tool, HiGeMine, our proposed framework for hierarchical genre classification, we observed a severe data imbalance issue in the training set. To address this, we performed data augmentation on both the blurbs and user reviews. We utilized the Gemini-2.5-flash [2] LLM in a controlled setting, applying prompt engineering to generate alternative textual variants that preserve the original semantic content, contextual coherence, and genre relevance. All generated samples were subsequently validated by human annotators to ensure quality. The augmented data were used exclusively during the training phase and were strictly excluded from validation and test sets. This augmentation strategy was intended to support model generalization and robustness without affecting evaluation outcomes. Table A.1 presents the distribution of the augmented data.

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TABLE A.1: Distribution of genre categories in the dataset

Genre ID	Genre name	Fiction		Non-fiction	
		Org.	Aug.	Org.	Aug.
1	Animals & Wildlife & Pets	542	648	183	598
2	Arts & Photography	1026	1180	483	674
3	Biographies & Memoir	-	-	1145	348
4	Business & Money	40	699	235	526
5	Children's Book	1525	1325	237	736
6	Comics & Graphic	328	316	92	516
7	Computers & Technology	25	420	75	546
8	Cookbooks & Food & Wine	69	763	167	618
9	Crafts & Hobbies & Home	-	-	43	566
10	Environment & Plant	84	537	256	548
11	Family & Parenting & Relationships	200	535	244	398
12	Fashion & Lifestyle	609	341	163	580
13	Health & Fitness & Dieting	30	476	269	576
14	History	1354	1183	1398	610
15	Humanities	441	433	1349	438
16	Humor & Entertainment	881	667	332	332
17	Literature	2183	1035	532	306
18	Medical	62	664	152	522
19	Meta Text	33	778	81	604
20	Mystery & Thriller & Suspense & Horror	1752	917	358	798
21	Mythology & Religion & Spirituality	707	884	654	394
22	Press & Media	130	471	145	612
23	Reference & Language	84	548	804	586
24	Romance	1171	677	70	590
25	Science & Math	349	252	548	428
26	Science Fiction & Fantasy	2257	835	-	-
27	Self-help & Motivation	66	700	590	414
28	Sports & Outdoors	153	483	127	606
29	Teen & Young Adult	1435	454	165	496
30	Travel	78	689	334	456
Genre Count		5612	12149	3763	11739

Org.: # Original, Aug.: # Augmented

APPENDIX B

IMPLEMENTATION DETAILS AND HYPERPARAMETER CONFIGURATION

All experiments were implemented in PyTorch 2.5.1 and conducted on a workstation equipped with an Intel Xeon G-6240R CPU, 192 GB RAM, and an NVIDIA A100 GPU with 80 GB memory.

Following the hierarchical design of HiGeMine, the LM encoders were first fine-tuned independently for Level-1 binary classification and Level-2 multi-label classification. The resulting encoders were subsequently frozen and used to initialize the remaining components of the framework, enabling stable and task-specific semantic representation learning during graph-based optimization.

All models were optimized using AdamW [3] with learning rates of 10^{-4} and 10^{-3} for Level-1 and Level-2, respectively. A dropout rate of 0.1 was applied throughout the framework to improve generalization. To mitigate overfitting, early stopping with a patience of 5 epochs based on validation performance was employed, with a maximum training budget of 200 epochs. The remaining hyperparameter settings are summarized in Table B.1. To ensure robustness and reproducibility, each experiment was conducted independently ten times, and the average performance across all runs is reported.

TABLE B.1: Brief details of hyperparameters

Attributes	Values
λ_1 & λ_2	[0, 1] & [0, 1]
LM Encoder Feature	768
GloVe Encoder Feature	Dimension: 300
Learnable Embedding Layer	300
GCN (B - T , P - T , and label co-occurrence network)	No. of layers: 2 Dimension: 128
Level-1 Dense Network	No. of layers: 2 Dimension: 128 & 2
Level-2 Dense Network	No. of layers: 1 Dimension: 128

APPENDIX C

REVIEW FILTERING: IMPACT OF SIMILARITY SCORE PERCENTILE

While we give the important observation related to the review filtering process in the main paper, here we provide the more experimental details. The review filtering process relies on using the pre-trained LM to yield filtered reviews that are more aligned with the blurb text. We clarify that the dynamic threshold (Ψ) used in the review filtering process (Section III-B of the main paper) is computed as $\Psi = \min(0.5, Q_{0.75}(d^i))$, where $Q_{0.75}(d^i)$ denotes the 75th percentile of similarity scores across all reviews of a book. This formulation enables adaptive review filtering without being overly strict.

TABLE C.1: Performance of HiGeMine with varying percentile of similarity score in the review filtering

Percentile	Level-1		Level-2					
			Fiction			Non-Fiction		
	$\mathcal{F} \uparrow$	$\mathcal{A}_c \uparrow$	$\mathcal{F}_\mu \uparrow$	$\mathcal{F}_m \uparrow$	$\mathcal{H}\mathcal{L} \downarrow$	$\mathcal{F}_\mu \uparrow$	$\mathcal{F}_m \uparrow$	$\mathcal{H}\mathcal{L} \downarrow$
100	94.71	95.80	66.87	55.07	0.0886	72.64	65.30	0.0676
90	95.57	96.48	80.45	72.50	0.0467	79.90	75.06	0.0457
75	96.91	97.50	81.89	77.24	0.0436	81.86	78.57	0.0401
60	96.84	97.44	81.49	76.40	0.0447	80.85	76.52	0.0432
45	96.84	97.44	81.07	74.02	0.0455	80.37	76.39	0.0443
30	96.77	97.39	81.25	73.00	0.0461	80.36	76.35	0.0446
15	96.64	97.28	80.37	72.53	0.0476	79.56	74.78	0.0468
0	96.50	97.17	79.60	70.85	0.0511	79.62	75.21	0.0458

Table C.1 reports the performance of HiGeMine under different similarity-score percentiles across both Level-1 and Level-2. The results indicate that moderate semantic filtering is important for effective hierarchical genre classification. While Level-1 performance remains relatively stable under noisy reviews, Level-2 multi-label prediction is considerably more sensitive to review quality. Overall, the 75th percentile provides the best balance between noise reduction and information preservation across both hierarchical levels.

APPENDIX D

DETAILED MISPREDICTION AND ERROR ANALYSIS

We posit that genre co-occurrence patterns play a major role in mispredictions. In this section, we analyze misclassifications arising either from indistinguishable phrasing within blurbs or reviews, or from frequent genre co-occurrence as follows:

Fig. D.1 presents a qualitative analysis of HiGeMine's classification errors caused by misleading genre information in blurbs and reviews that trigger systematic prediction inaccuracies across all model configurations, HiGeMine (1, 1), HiGeMine (0, 0), HiGeMine (0.5, 0.5), and standard HiGeMine. The segments

Dark Horse (ISBN:9780380973149)

Blurb

Believing that he will have no trouble being nominated for congress after **the death of a long-time political incumbent**, lawyer Mitch Dutton finds a **terrifying** challenge in dark **horse** Shakespeare McCann, who is out to win at any cost.

When I picked up this book and saw that it had been written by a successful screenwriter I thought that as a person for whom scenarios, characters and words are his livelihood then authorship would be something that Mr. Richardson should be adept at. I was horribly wrong. The plot lines were at best implausible and at worst downright ridiculous. Holes appear within these plots leaving the reader to wonder how or why sets of circumstances came together to create even the possibility that the scene being presented could have occurred. Had the author written a story about shape-shifting **lizards** from the planet Zog attempting to take over the Earth by initially taking control of the American government by, at first, having a congressman from a remote southern Texan town then **the reader can potentially enter into the fantasy and ride with it**. This novel lacked even that level of credibility. Just very poor.

DNF at about 50%. Literally so bad. I'm annoyed that I spent so long trying to stick it out and finish it, but I simply cannot read any more. Men, if you are going to put women in your books purely so they can be raped, cheated on, or being the object of the main character's cheating, maybe just don't put them in at all.

Crusading Texas Lawyer Mitchell Dutton is running for the U.S. Congress – and is a virtual shoo-in come November. But his opponent has different plans. The opposition nominee, Shakespeare "Shakes" McCann, is an almost total unknown – a dark horse candidate with no past and no principles who has come out of nowhere to challenge Mitch Dutton. There is no weakness McCann will not exploit, no secret he cannot expose. He will lie to reporters and newscasters, alter photographs and circulate fraudulent documents. If all else fails, he will resort to violence. And no one is safe – not Dutton's friends, his colleagues, his beautiful vulnerable wife...not even the leaders of McCann's own party. There is nothing the dark horse won't do to get elected. Now front-runner Dutton is running scared. Because Shakespeare McCann would kill to be Congressman....(cover blurb) **The good news: it's original, frightening, cynical and even funny;** it moves along at lightning speed; and I couldn't wait to find out what would happen next. The bad news: Omigosh, you wouldn't believe how badly written this thing is: sentence fragments, trite phrasing, literary clichés, astonishing coincidences and some nearly unbelievable plot twists....Still, the author's inventiveness and imagination put him in good stead. Even as poorly executed as this novel is, I felt compelled to keep reading. If Richardson actually learns to write, he'll be unstoppable. A timely read in an election year. Recommended, with reservations.

Lots of twists and turns. Different

Good

Actual Genre: Fiction → Humanities

HiGeMine ($\lambda_1 = 1, \lambda_2 = 1$): Fiction → Animals & Wild life & Pets, Mystery & Thriller & Suspense & Horror

HiGeMine ($\lambda_1 = 0, \lambda_2 = 0$): Fiction → Fashion & Lifestyle, Humor & Entertainment, Mystery & Thriller & Suspense & Horror, Science Fiction & Fantasy

HiGeMine ($\lambda_1 = 0.5, \lambda_2 = 0.5$): Fiction → Animals & Wild life & Pets, Fashion & Lifestyle, Mystery & Thriller & Suspense & Horror

HiGeMine: Fiction → Humor & Animals & Wild life & Pets, Mystery & Thriller & Suspense & Horror

Fig. D.1: Qualitative analysis for misprediction: misleading genre indicative phrases in blurbs and reviews (highlighted in blue) may trigger systematic prediction inaccuracies across all variation of HeGeMine.

marked in blue show how the specific genre-indicative terms in both modalities result in consistent misclassifications despite the varying modality ratios of the model. This underscores the need for improved noise detection mechanisms and context-sensitive feature weighting to enhance the robustness of the model against adversarial textual patterns.

Table D.1 (a) and Table D.2 (a) present pairwise genre co-occurrence counts in the training set corresponding to *fiction* (F) and *non-fiction* (NF), where cell (ID_i, ID_j) represents samples containing both genre ID_i and ID_j . Table D.1 (b) and Table D.2 (b) show co-occurrence ratios between genres, quantifying how frequently genre ID_i appears given genre ID_j . Finally, Table D.1 (c) and Table D.2 (c) report the misprediction rate of genre ID_i misclassified as genre ID_j .

Our analysis from Table D.1 and Table D.2 highlights that frequent co-occurrence between the two genres, coupled with the significant differences in the number of samples in one genre without association with another, causes a high misprediction rate. For instance, in Table D.2 (a) NF: *craft & hobbies & home* (genre ID: 9) and NF: *reference & language* (genre ID: 23) appeared in 26 samples together. However, we have a total of 32 samples associated with genre ID: 9, which leaves only six genre ID: 9 and non-ID: 23 samples. Conversely, we have a total of 567 samples associated with genre ID: 23, leaving 541 samples of genre ID: 23 and non-ID: 9. It is creating an imbalance between genres ID: 9 and non-ID: 23 and non-ID: 9 ID: 23, leading to classify genre ID: 9 to genre ID: 23, with the misclassification rate of one (see Fig. D.2). The traditional data

augmentation methods are limited in their ability to address this imbalance issue, as increasing the number of samples for genre ID: 9 genre also increases the genre ID: 23 samples due to the inter-genre relation. Furthermore, in Table D.1, we observe a similar issue on genre ID pairs: (30 & 17), (15 & 17), and (19 & 20).

APPENDIX E LIMITATIONS

Although HiGeMine demonstrates strong performance in hierarchical book genre classification compared to baseline methods, it exhibits several limitations related to language coverage, modality usage, and hierarchical modeling. In this section, we discuss these limitations and outline potential directions for future work.

(i) *Low-Resource and Multilingual Settings*: HiGeMine relies on pretrained language models and encoder-based semantic alignment to estimate semantic consistency between blurbs and crowd-sourced reviews. Consequently, the effectiveness of the review refinement module depends on the quality of the underlying contextual representations. In low-resource or multilingual settings, pretrained encoders may exhibit weaker semantic grounding due to limited training corpora, reduced vocabulary coverage, or insufficient exposure to culture-specific literary expressions. As a result, semantically relevant reviews may receive inaccurate similarity estimates and be incorrectly filtered during the refinement stage. Furthermore, genre semantics are often expressed through implicit narratives,

Yankee Magazine's Vinegar, Duct Tape, Milk Jugs & More (ISBN:9780899093796)

Blurb

From milk jugs to egg cartons to mayonnaise and more, your home contains an untapped gold mine of materials that you can use to solve problems, fix what breaks, or create substitutes for things you'd otherwise have to buy. The trick is figuring out what among these objects is treasure and what's trash. That's where Earl Proulx and the editors of Yankee magazine can help. Drawing on their own creative ideas—and those of ingenious Yankees all over New England—they've come up with more than 1,000 clever ways to put common household objects to uncommon and valuable uses. The result is a book that will benefit you in five clear ways. 1. This book will empower you. Other people might be stymied when, say, Spot knocks a glass of grape juice on that elegant white rug. Not you. You'll be able to lift that stain yourself—without buying some expensive remover—just by applying a dab of shaving cream. Vinegar, Duct Tape, Milk Jugs, and More contains dozens of clever do-it-yourself ideas like this. ...

Unlike the last "helpful" household tips books I read (1,628 Country Shortcuts: From 1,628 Country People), this one is actually good and, yes, helpful. It even has an index, which makes me really happy. With an index, I can actually find a solution when I need one. Also helpful is a ten page list at the front of the book of the author's "very best out-of-the-ordinary uses for ordinary items around the house. So if you want to know what to do with old jeans or powdered milk or soda bottles, just consult the list for ideas. The tip I like best so far is the suggestion to use a game board from a no longer played board game as a lap desk. I have access to an old Scrabble board that I plan to cover in clear contact paper (as soon as I find some cheap clear contact paper at the thrift store). Once it's covered, it will be easy to wipe glue off. ...

OK, first of all, this is not a good book to give to a hoarder. The author saves every can, bottle, egg carton and milk jug he ever purchased. I can't really say that I will do anything suggested ...

Nice little book all about household hints. Easy read - some good hints, some not so good and some crazy! As stated on the book cover - "From the attic to the basement; From the kitchen to the bath; From the home to the office; From the garden to the garage." Covers it all - plus - recreation, gift giving, sports, hobbies, day trips, travel, weekends, celebrations and pets. And OMG the uses for a gallon milk jug - endless I tell you - endless! Quick effortless read.

This is kinda like the dictionary for those of us who look up a word and keep on reading.... I needed some information on cleaning a shower stall without using chemicals and BAM! ...

Informative book about how to improvise. It reminded me of my childhood years growing up in a poor household and having to make-do with what we had. However, I will say it is a bit dated and risky, like kitty litter face mask back then it was just clay (maybe), nowadays I wouldn't dare, too many chemicals are in everything. But to be fair there was a disclaimer in the beginning of the book to advise your own caution. Take what you will with this review.

An impressive set of uses for vinegar and baking soda.... Earl's main message also seems to be throw nothing away and everything has a new use. Entertaining and enlightening, really.

A lot of good tips in this book however I would not condone cleaning with ammonia.

Awesome...

This book is great, it is full of great suggestions on how to make fun things, and all those things listed above. Clean things. Also gives great ideas on how to reuse things normally just tossed away. I made a bird feeder out of a milk-jug and the birds love it! ...

Good idea book with new suggestions for old items you already have. Great for people who want to learn to use or recycle items in new ways. Some of the ideas I already knew. All in all a good book. Worth owning if you want to use items instead of throwing them out to end up in the landfill.

Actual Genre: Non-fiction → Crafts & Hobbies & Home, Reference & Language, Self-Help & Motivation

HiGeMine ($\lambda_1 = 1, \lambda_2 = 1$): Non-fiction → Arts & Photography, Humor & Entertainment, Press & Media, Reference & Language

HiGeMine ($\lambda_1 = 0, \lambda_2 = 0$): Non-fiction → Environment & Plant, Reference & Language

HiGeMine ($\lambda_1 = 0.5, \lambda_2 = 0.5$): Non-fiction → Reference & Language

HiGeMine: Non-fiction → Reference & Language

Fig. D.2: Qualitative analysis for misprediction due to genre indistinguishability

idiomatic phrases, culturally grounded references, or region-specific terminology, which may not align consistently across languages and domains. Such distributional shifts can reduce the reliability of cross-document semantic alignment and hierarchical genre inference. Future work will therefore explore multilingual and cross-lingual representation learning, language-adaptive semantic alignment, and domain-specific instruction tuning to improve robustness under linguistically diverse and low-resource literary settings.

(ii) *Lack of Multi-modal Data Integration*: HiGeMine currently relies exclusively on textual evidence derived from book blurbs and user reviews, without incorporating additional modalities such as cover images, author metadata, publication information, or reader interaction statistics. However, literary genres are often expressed through complementary multi-modal cues. For example, cover design frequently encodes visual genre priors through typography, color composition, and artwork style, while metadata such as author identity, publication category, or reader engagement patterns may provide additional contextual

signals for genre disambiguation. The absence of such modalities may limit the framework's robustness in scenarios where textual evidence is sparse, ambiguous, or semantically noisy, particularly for cold-start books with limited reviews or minimal blurbs. Furthermore, certain visually distinctive genres may benefit from joint visual-semantic reasoning beyond textual representations alone. Future work will therefore explore multi-modal extensions of HiGeMine through modality-aware fusion strategies, cross-modal semantic alignment, and graph-based heterogeneous representation learning integrating textual, visual, and structured metadata signals within a unified hierarchical reasoning framework.

(iii) *Limited Support for Deeper Hierarchies*: HiGeMine currently assumes a fixed two-level hierarchy consisting of coarse-grained *fiction/non-fiction* categorization followed by fine-grained genre prediction. While effective for coarse-to-fine literary reasoning, real-world genre taxonomies are often deeper and more dynamic, containing nested subgenres and complex parent-child relations. The current framework does

not explicitly model multi-hop hierarchical dependencies across arbitrary hierarchy depths, which may limit scalability to more complex taxonomic structures. Future work will therefore explore multi-level hierarchical message passing and taxonomy-aware graph reasoning for deeper and more flexible literary genre ontologies.

(iv) *Single-Source Review Data*: HiGeMine is currently evaluated using Goodreads reviews, whose literary-focused community typically produces long, theme-centric reviews that align closely with publisher-authored blurbs. This stylistic consistency likely benefits the proposed semantic refinement module. In contrast, platforms such as Amazon often contain shorter, purchase-oriented, and opinion-driven reviews, introducing distributional shifts in vocabulary, semantic density, and genre expression that may affect semantic alignment quality. Consequently, the current framework may exhibit reduced robustness under cross-platform review distributions. Future work will therefore extend the benchmark using multi-platform review sources to evaluate cross-domain generalization and robustness of the proposed semantic refinement framework.

REFERENCES

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TABLE D.1: Fiction genre misprediction analysis

(a) Sample Count																																	
ID ↓→	1	2	4	5	6	7	8	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30					
1	399	176	0	331	8	0	15	23	34	0	5	21	9	52	152	2	2	94	17	0	0	8	4	139	3	0	144	0					
2	176	748	1	419	11	4	29	20	69	17	7	49	12	149	261	3	3	125	45	4	15	63	11	381	7	10	172	5					
4	0	1	28	0	1	1	0	0	0	0	0	6	5	2	11	0	0	13	2	1	0	0	0	4	4	0	0	1					
5	331	419	0	1119	31	3	26	30	85	0	10	150	20	199	458	6	6	269	109	3	7	44	2	465	8	33	581	3					
6	8	11	1	31	222	1	1	0	0	0	0	23	16	92	8	0	0	53	0	3	1	26	3	103	0	1	20	0					
7	0	4	1	3	1	18	0	0	0	0	0	1	1	2	6	0	0	5	0	0	1	1	1	16	0	6	1	1					
8	15	29	0	26	1	0	55	2	4	5	4	2	3	16	27	0	0	11	2	1	0	9	2	11	0	0	3	3					
10	23	20	0	30	0	0	2	70	5	1	0	14	11	5	43	0	0	15	8	0	0	4	5	21	0	4	8	3					
11	34	69	0	85	0	0	4	5	150	14	2	37	0	20	93	4	0	5	4	0	0	25	0	9	3	0	29	1					
12	0	17	0	0	0	0	5	1	14	434	3	121	71	45	205	5	2	125	15	0	1	181	0	100	9	0	12	8					
13	5	7	0	10	0	0	4	0	2	3	23	1	6	3	13	8	0	4	1	0	0	0	0	5	0	0	8	0					
14	21	49	6	150	23	1	2	14	37	121	1	928	66	37	303	11	5	252	159	17	1	263	31	261	13	6	237	20					
15	9	12	5	20	16	1	3	11	0	71	6	66	324	37	210	12	2	70	45	0	4	39	1	106	14	2	32	9					
16	52	149	2	199	92	2	16	5	20	45	3	37	37	624	251	3	7	116	10	10	6	59	16	204	3	7	114	6					
17	152	261	11	458	8	6	27	43	93	205	13	303	210	251	1565	25	9	395	176	10	29	333	89	502	22	16	443	40					
18	2	3	0	6	0	0	0	4	4	5	8	11	12	3	25	51	0	14	2	2	0	10	3	10	0	1	8	0					
19	2	3	0	6	0	0	0	0	0	2	0	5	2	7	9	0	23	6	1	0	3	2	1	9	0	0	4	0					
20	94	125	13	269	53	5	11	15	5	125	4	252	70	116	395	14	6	1250	109	43	0	214	85	534	6	34	309	26					
21	17	45	2	109	0	0	2	8	4	15	1	159	45	10	176	2	1	109	488	0	7	121	8	182	43	0	94	3					
22	0	4	1	3	3	0	1	0	0	0	0	17	0	10	10	2	0	43	0	92	2	0	4	69	0	0	0	1					
23	0	15	0	7	1	1	0	0	0	1	0	1	4	6	29	0	3	0	7	2	63	0	9	46	0	6	2	2					
24	8	63	0	44	26	1	9	4	25	181	0	263	39	59	333	10	2	214	121	0	0	795	30	219	14	3	150	6					
25	4	11	0	2	3	1	2	5	0	0	0	31	1	16	89	3	1	85	8	4	9	30	226	130	0	0	3	0					
26	139	381	4	465	103	16	11	21	9	100	5	261	106	204	502	10	9	534	182	69	46	219	130	1597	8	42	478	6					
27	3	7	4	8	0	0	0	0	3	9	0	13	14	3	22	0	0	6	43	0	0	14	0	8	56	0	6	1					
28	0	10	0	33	1	6	0	4	0	0	0	6	2	7	16	1	0	34	0	0	6	3	0	42	0	100	24	0					
29	144	172	0	581	20	1	3	8	29	12	8	237	32	114	443	8	4	309	94	0	2	150	3	478	6	24	1021	0					
30	0	5	1	3	0	1	3	3	1	8	0	20	9	6	40	0	0	26	3	1	2	6	0	6	1	0	0	58					

(b) Co-occurrence																																	
ID ↓→	1	2	4	5	6	7	8	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30					
1	1.00	0.44	0.00	0.83	0.02	0.00	0.04	0.06	0.09	0.00	0.01	0.05	0.02	0.13	0.38	0.01	0.01	0.24	0.04	0.00	0.00	0.02	0.01	0.35	0.01	0.00	0.36	0.00					
2	0.24	1.00	0.00	0.56	0.01	0.01	0.04	0.03	0.09	0.02	0.01	0.07	0.02	0.20	0.35	0.00	0.00	0.17	0.06	0.01	0.02	0.08	0.01	0.51	0.01	0.01	0.23	0.01					
4	0.00	0.04	1.00	0.00	0.04	0.04	0.00	0.00	0.00	0.00	0.00	0.21	0.18	0.07	0.39	0.00	0.00	0.46	0.07	0.04	0.00	0.00	0.14	0.14	0.00	0.00	0.04						
5	0.30	0.37	0.00	1.00	0.03	0.00	0.02	0.03	0.08	0.00	0.01	0.13	0.02	0.18	0.41	0.01	0.01	0.24	0.10	0.00	0.01	0.04	0.00	0.42	0.01	0.03	0.52	0.00					
6	0.04	0.05	0.00	0.14	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.10	0.07	0.41	0.04	0.00	0.00	0.24	0.00	0.01	0.00	0.12	0.01	0.46	0.00	0.00	0.09	0.00					
7	0.00	0.22	0.06	0.17	0.06	1.00	0.00	0.00	0.00	0.00	0.00	0.06	0.06	0.11	0.33	0.00	0.00	0.28	0.00	0.00	0.06	0.06	0.06	0.89	0.00	0.33	0.06	0.06					
8	0.27	0.53	0.00	0.47	0.02	0.00	1.00	0.04	0.07	0.09	0.07	0.04	0.05	0.29	0.49	0.00	0.00	0.20	0.04	0.02	0.00	0.16	0.04	0.20	0.00	0.00	0.05	0.05					
10	0.33	0.29	0.00	0.43	0.00	0.00	0.03	1.00	0.07	0.01	0.00	0.20	0.16	0.07	0.61	0.00	0.00	0.21	0.11	0.00	0.00	0.06	0.07	0.30	0.00	0.06	0.11	0.04					
11	0.23	0.46	0.00	0.57	0.00	0.00	0.03	0.03	1.00	0.09	0.01	0.25	0.00	0.13	0.62	0.03	0.00	0.03	0.03	0.00	0.17	0.00	0.06	0.02	0.00	0.19	0.01						
12	0.00	0.04	0.00	0.00	0.00	0.00	0.01	0.00	0.03	1.00	0.01	0.28	0.16	0.10	0.47	0.01	0.00	0.29	0.03	0.00	0.00	0.42	0.00	0.23	0.02	0.00	0.03	0.02					
13	0.22	0.30	0.00	0.43	0.00	0.00	0.17	0.00	0.09	0.13	1.00	0.04	0.26	0.13	0.57	0.35	0.00	0.17	0.04	0.00	0.00	0.00	0.00	0.22	0.00	0.00	0.35	0.00					
14	0.02	0.05	0.01	0.16	0.02	0.00	0.00	0.02	0.04	0.13	0.00	1.00	0.07	0.04	0.33	0.01	0.01	0.27	0.17	0.02	0.00	0.28	0.03	0.28	0.01	0.01	0.26	0.02					
15	0.03	0.04	0.02	0.06	0.05	0.00	0.01	0.03	0.00	0.22	0.02	0.20	1.00	0.11	0.65	0.04	0.01	0.22	0.14	0.00	0.01	0.12	0.00	0.33	0.04	0.01	0.10	0.03					
16	0.08	0.24	0.00	0.32	0.15	0.00	0.03	0.01	0.03	0.07	0.00	0.06	0.06	1.00	0.40	0.00	0.01	0.19	0.02	0.02	0.01	0.09	0.03	0.33	0.00	0.01	0.18	0.01					
17	0.10	0.17	0.01	0.29	0.01	0.00	0.02	0.03	0.06	0.13	0.01	0.19	0.13	0.16	1.00	0.02	0.01	0.25	0.11	0.01	0.02	0.21	0.06	0.32	0.01	0.01	0.28	0.03					
18	0.04	0.06	0.00	0.12	0.00	0.00	0.00	0.00	0.08	0.10	0.16	0.22	0.24	0.06	0.49	1.00	0.00	0.27	0.04	0.04	0.00	0.20	0.06	0.20	0.00	0.02	0.16	0.00					
19	0.09	0.13	0.00	0.26	0.00	0.00	0.00	0.00	0.00	0.09	0.00	0.22	0.09	0.30	0.39	0.00	1.00	0.26	0.04	0.00	0.13	0.09	0.04	0.39	0.00	0.00	0.17	0.00					
20	0.08	0.10	0.01	0.22	0.04	0.00	0.01	0.01	0.00	0.10	0.00	0.20	0.06	0.09																			

TABLE D.2: Non-fiction genre misprediction analysis

(a) Sample Count																														
ID ↓→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	27	28	29	30	
1	149	27	38	0	36	4	0	4	0	60	2	0	4	24	14	22	11	2	0	0	4	0	20	0	74	2	0	7	6	
2	27	360	107	2	67	21	6	0	11	12	6	5	2	161	69	36	33	4	6	10	27	39	85	8	46	2	1	26	9	
3	38	107	795	31	27	17	7	29	2	48	14	35	8	378	174	112	159	21	24	123	80	45	21	10	43	22	51	25	147	
4	0	2	31	161	0	5	19	0	0	7	0	0	46	84	4	2	0	0	0	0	14	8	33	0	13	81	0	2	0	
5	36	67	27	0	174	5	0	2	2	19	39	0	7	45	33	13	10	6	5	0	1	0	31	0	66	17	0	50	0	
6	4	21	17	5	5	77	0	0	1	2	1	5	1	29	27	31	6	0	2	3	2	1	18	1	20	2	0	2	1	
7	0	6	7	19	0	0	50	0	0	2	0	0	0	17	20	2	0	1	1	0	0	0	9	0	23	3	2	0	0	
8	4	0	29	0	2	0	0	116	0	16	0	2	44	11	7	8	12	3	1	0	2	0	40	0	8	18	0	2	23	
9	0	11	2	0	2	1	0	0	32	2	0	7	0	7	4	1	1	0	4	0	1	0	26	0	5	1	0	0	0	
10	60	12	48	0	19	2	2	16	2	185	0	0	13	52	36	7	11	2	0	14	8	2	35	0	85	1	9	6	24	
11	2	6	14	7	39	1	0	0	0	165	9	24	4	95	9	3	16	9	0	33	0	35	13	9	83	2	14	0	0	
12	0	5	35	0	0	5	0	2	7	0	9	101	0	28	46	12	25	2	6	11	7	0	18	16	5	16	3	0	2	
13	4	2	8	0	7	1	0	44	0	13	24	0	176	1	66	0	1	46	0	0	21	0	76	1	30	109	5	0	0	
14	24	161	378	46	45	29	17	11	7	52	4	28	1	992	340	58	181	16	24	147	163	59	138	12	156	2	21	43	120	
15	14	69	174	84	33	27	20	7	4	36	95	46	66	340	919	58	159	41	26	42	208	30	179	23	140	210	6	49	26	
16	22	36	112	4	13	31	2	8	1	7	9	12	0	58	58	232	23	0	2	9	4	12	32	1	9	7	10	6	35	
17	11	33	159	2	10	6	0	12	1	11	3	25	1	181	159	23	384	3	18	26	56	18	52	16	9	1	0	17	52	
18	2	4	21	0	6	0	1	3	0	2	16	2	46	16	41	0	3	98	0	6	15	2	24	2	36	39	0	0	0	
19	0	6	24	0	5	2	1	1	4	0	9	6	0	24	26	2	18	0	63	4	6	0	29	0	4	3	0	3	2	
20	0	10	123	0	0	3	0	0	0	14	0	11	0	147	42	9	26	6	4	246	21	20	19	8	14	2	18	0	70	
21	4	27	80	14	1	2	0	2	1	8	33	7	21	163	208	4	56	15	6	21	449	0	68	14	42	125	1	0	14	
22	0	39	45	8	0	1	0	0	0	2	0	0	0	59	30	12	18	2	0	20	0	109	15	1	10	0	1	2	10	
23	20	85	21	33	31	18	9	40	26	35	35	18	76	138	179	32	52	24	29	19	68	15	567	11	65	116	26	28	16	
24	0	8	10	0	0	1	0	0	0	0	13	16	1	12	23	1	16	2	0	8	14	1	11	54	5	16	1	0	1	
25	74	46	43	13	66	20	23	8	0	85	9	5	30	156	140	9	9	36	4	14	42	10	65	5	388	34	0	26	12	
27	2	2	22	81	17	2	3	18	5	1	83	16	109	2	210	7	1	39	3	2	125	0	116	16	34	397	6	4	2	
28	0	1	51	0	0	0	2	0	1	9	2	3	5	21	6	10	0	0	0	18	1	1	26	1	0	6	91	0	17	
29	7	26	25	2	50	2	0	2	0	6	14	0	0	43	49	6	17	0	3	0	0	2	28	0	26	4	0	113	0	
30	6	9	147	0	0	1	0	23	0	24	0	2	0	120	26	35	52	0	2	70	14	10	16	1	12	2	17	0	236	
(b) Co-occurrence																														
ID ↓→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	27	28	29	30	
1	1.00	0.18	0.26	0.00	0.24	0.03	0.00	0.03	0.00	0.40	0.01	0.00	0.03	0.16	0.09	0.15	0.07	0.01	0.00	0.00	0.03	0.00	0.13	0.00	0.50	0.01	0.00	0.05	0.04	
2	0.08	1.00	0.30	0.01	0.19	0.06	0.02	0.00	0.03	0.03	0.02	0.01	0.01	0.45	0.19	0.10	0.09	0.01	0.02	0.03	0.08	0.11	0.24	0.02	0.13	0.01	0.00	0.07	0.03	
3	0.05	0.13	1.00	0.04	0.03	0.02	0.01	0.04	0.00	0.06	0.02	0.04	0.01	0.48	0.22	0.14	0.20	0.03	0.03	0.15	0.10	0.06	0.03	0.01	0.05	0.03	0.06	0.03	0.18	
4	0.00	0.01	0.19	1.00	0.00	0.03	0.12	0.00	0.00	0.04	0.00	0.00	0.00	0.29	0.52	0.02	0.01	0.00	0.00	0.00	0.09	0.05	0.20	0.00	0.08	0.50	0.00	0.01	0.00	
5	0.21	0.39	0.16	0.00	1.00	0.03	0.00	0.01	0.01	0.11	0.22	0.00	0.04	0.26	0.19	0.07	0.06	0.03	0.03	0.00	0.01	0.00	0.18	0.00	0.38	0.10	0.00	0.29	0.00	
6	0.05	0.27	0.22	0.06	0.06	1.00	0.00	0.00	0.01	0.03	0.01	0.06	0.01	0.38	0.35	0.40	0.08	0.00	0.03	0.04	0.03	0.01	0.23	0.01	0.26	0.03	0.00	0.03	0.01	
7	0.00	0.12	0.14	0.38	0.00	0.00	1.00	0.00	0.04	0.00	0.00	0.00	0.00	0.34	0.40	0.04	0.00	0.02	0.02	0.00	0.00	0.00	0.18	0.00	0.46	0.06	0.04	0.00	0.00	
8	0.03	0.00	0.25	0.00	0.02	0.00	0.00	1.00	0.00	0.14	0.00	0.02	0.38	0.09	0.06	0.07	0.10	0.03	0.01	0.00	0.02	0.00	0.34	0.00	0.07	0.16	0.00	0.02	0.20	
9	0.00	0.34	0.06	0.00	0.06	0.03	0.00	0.00	1.00	0.06	0.00	0.22	0.00	0.22	0.13	0.03	0.03	0.00	0.13	0.00	0.03	0.00	0.81	0.00	0.00	0.16	0.03	0.00	0.00	
10	0.32	0.06	0.26	0.00	0.10	0.01	0.01	0.09	0.01	1.00	0.00	0.00	0.07	0.28	0.19	0.04	0.06	0.01	0.00	0.08	0.04	0.01	0.19	0.00	0.46	0.01	0.05	0.03	0.13	
11	0.01	0.04	0.08	0.04	0.24	0.01	0.00	0.00	0.00	0.00	1.00	0.05	0.15	0.02	0.58	0.05	0.02	0.10	0.05	0.00	0.20	0.00	0.21	0.08	0.05	0.50	0.01	0.08	0.00	
12	0.00	0.05	0.35	0.00	0.00	0.05	0.00	0.02	0.07	0.00	0.09	1.00	0.00	0.28	0.46	0.12	0.25	0.02	0.06	0.11	0.07	0.00	0.18	0.16	0.05	0.16	0.03	0.00	0.02	
13	0.02	0.01	0.05	0.00	0.04	0.01	0.00	0.25	0.00	0.07	0.14	0.00	1.00	0.01	0.38	0.00	0.01	0.26	0.00	0.00	0.12	0.00	0.43	0.01	0.17	0.62	0.03	0.00	0.00	
14	0.02	0.16	0.38	0.05	0.05	0.03	0.02	0.01	0.01	0.05	0.00	0.03	0.00	1.00	0.34	0.06	0.18	0.02	0.02	0.15	0.16	0.06	0.14	0.01	0.16	0.00	0.02	0.04	0.12	
15	0.02	0.08	0.19	0.09	0.04	0.03	0.02	0.01	0.00	0.04	0.10	0.05	0.07	0.37	1.00	0.06	0.17	0.04	0.03	0.05	0.23	0.03	0.19	0.03	0.15	0.23	0.01	0.05	0.03	
16	0.09	0.16	0.48	0.02	0.06	0.13	0.01	0.03	0.00	0.03	0.04	0.05	0.00	0.25	0.25	1.00	0.10													